Electronic Design Lab

Course Structure: 1 - 1 - 4 - 4
Course Code:
Prerequisites: None

Course Objective: This module equips the students to learn different CAD, EDA tools & HDL for hardware and be conversant with the measuring instruments.

Course Outcome

Having successfully completed this module the student will be able to use function generators, oscilloscopes and complex devices such as logic analysers, spectrum analysers, understand data sheets, debug HW and SW systems, design simple digital ICs and perform synthesis of simple electronic circuits and systems.

Syllabus

- Effective use of laboratory equipment and Synthesis vs analysis
- Effective use of design resources, Cadence ICFB, Matlab, Spice, ModelSim, Xilinx ISE
- Manufacturing techniques, fabless design, EMC
- Learn HDL, System Verilog

The course has two experiments

1. Build a part of an 8-bit Microprocessor using Discrete components and verify them
2. Verify Various Analog Circuits using LM741

Resources

- Web Resources, Datasheets